

MIRA JÜRGENS

PhD Researcher | Uncertainty Quantification in Machine Learning

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 Ghent, Belgium

 Github

 Google Scholar

 LinkedIn

RESEARCH PROFILE

I am interested in both theoretical foundations as well as real-world applications of machine learning and statistics. Throughout my PhD, I worked on frameworks for evaluating epistemic uncertainty representations in machine learning, and applied uncertainty-aware prediction pipelines to real-world problems. I thrive in collaborative, interdisciplinary settings.

RESEARCH & PROFESSIONAL EXPERIENCE

Doctoral Researcher

Ghent University – Dept. of Data Analysis & Mathematical Modeling

Dec 2022 – present

Ghent, Belgium

Supervisor: Prof. Willem Waegeman. Key contributions include:

- ❖ Development of a nonparametric calibration test for credal set representations of epistemic uncertainty. Published in *Machine Learning* (Springer, 2025).
- ❖ Critical theoretical analysis of the evidential deep learning framework, exposing identifiability and convergence issues. Published at *ICML 2024*.
- ❖ Position paper about fundamental flaws in current uncertainty quantification methods due to bias contamination and partial variance capture (equal contribution). Currently under review.
- ❖ Adaption of a selective prediction framework for molecular structure retrieval in metabolomics. Currently under review.
- ❖ Joint work on application of conformal prediction to drug–target interaction uncertainty estimation, published at *COPA 2025*.
- ❖ Supervised 2 Master’s theses on uncertainty-related topics.
- ❖ Gave tutorials and supervised projects for the Master’s course *Predictive Modeling* (2023–2026).

Research Assistant

Fraunhofer Institute for Energy Economics and Energy System Technology (IEE)

Jan 2022 – Nov 2022

Kassel, Germany

Key contributions:

- ❖ Development of deep generative models for synthesizing time series in energy system analysis. Design of evaluation metrics and data pipelines. Collaboration in an industry-based project.
- ❖ Lead of acquisition and project management for an internal research initiative on GANs for renewable energy.
- ❖ Contribution to cross-departmental projects on adversarial robustness in energy systems. Writing of research proposals. Supervised student assistants.

Student Assistant & Master’s Thesis

Fraunhofer IEE

May 2021 – Feb 2022

Kassel, Germany

Thesis on GANs for synthetic wind power time series.

Earlier Experience

-  Hamburg | **Research Intern**, University of Hamburg, Institute for Oceanography (2019). Statistical modeling of energy transfer processes at smaller scales.
-  Göttingen | **Student Teaching Assistant**, University of Göttingen (2017–2021). Tutor for the lectures *Stochastics*, *Mathematics for Economists*, *Mathematics for Physicists*.

EDUCATION

Ph.D. Mathematical Modeling

Ghent University

Dec 2022 – Dec 2026

Ghent, Belgium

Expected completion: Nov 2026

M.Sc. Mathematics

University of Göttingen

Oct 2019 – Jan 2022

Göttingen, Germany

Grade: 1.5 (high distinction)

Profile: Mathematical Data Science

B.Sc. Mathematics

University of Göttingen

Oct 2015 – Mar 2019

Göttingen, Germany

Grade: 2.1

Major: Mathematical Stochastics

SKILLS

Research Methods

- Uncertainty Quantification
- Conformal Prediction
- Selective Prediction
- Calibration
- Bayesian Methods
- Deep Generative Models
- Statistical Analysis

Technical

- Python
- PyTorch
- R
- Git
- CI/CD
- MLflow
- Linux
- W&B
- LaTeX

Transferable

- Academic Writing
- Project Management
- Teaching & Supervision
- Peer Review
- Interdisciplinary Collaboration
- Scientific Communication

LANGUAGES

- ❖ **German** – Native
- ❖ **English** – Fluent
- ❖ **Dutch** – B2
- ❖ **Spanish** – A2

TALKS & PRESENTATIONS

Invited Talks

- ❖ **Heidelberg University** – Fundamental limitations of aleatoric and epistemic uncertainty disentanglement.  Heidelberg, Jan 2026
- ❖ **Amadeus** (industry) – Calibration testing for set-based uncertainty representations. Nov 2025, remote
- ❖ **KU Leuven** – Evidential deep learning and the faithfulness of epistemic uncertainty representations.  Leuven, Oct 2024

Contributed Talks & Posters

- ❖ **Workshop on Uncertainty in ML (WUML 2026)** – Contributed talk on selective prediction for molecular structure retrieval.  Tartu, Feb 2026
- ❖ **ECMLPKDD 2025** – Contributed talk: “A calibration test for evaluating set-based epistemic uncertainty representations”  Porto, Sep 2025
- ❖ **ICML 2024** – Poster: “Is Epistemic Uncertainty faithfully represented by Evidential Deep Learning methods?”  Vienna, Jul 2024
- ❖ **Workshop on Uncertainty in ML (WUML 2024)** – Contributed talk on pitfalls of evidential deep learning.  Munich, Feb 2024
- ❖ **Flanders AI Research Day 2024** – PhD pitch on uncertainty quantification in machine learning.  Ghent, Oct 2024

PUBLICATIONS

- [1] M. Jürgens, T. Mortier, E. Hüllermeier, V. Bengs, and W. Waegeman, “A calibration test for evaluating set-based epistemic uncertainty representations”, *Machine Learning* (Springer), vol. 114, no. 9, 2025.
- [2] M. Jürgens, N. Meinert, V. Bengs, E. Hüllermeier, and W. Waegeman, “Is Epistemic Uncertainty faithfully represented by Evidential Deep Learning methods?” in Proc. *ICML*, 41st International Conference on Machine Learning, 2024.
- [3] S. Jiménez*, M. Jürgens*, W. Waegeman, “Why machine learning models fail to fully capture epistemic uncertainty”, arXiv preprint, 2025.
[* equal contribution; **position paper**]
- [4] M. Jürgens, G. De Waele, M. Rakhshaninejad, and W. Waegeman, “When should we trust the annotation? Selective prediction for molecular structure retrieval from mass spectra”, arXiv preprint, 2026.
- [5] M. Rakhshaninejad, M. Jürgens, N. Dewolf, and W. Waegeman, “Conformal Prediction for Uncertainty Estimation in Drug-Target Interaction Prediction,” in *COPA*, Fourteenth Symposium on Conformal and Probabilistic Prediction with Applications, 2025.

PROFESSIONAL DEVELOPMENT & ACADEMIC SERVICE

Summer Schools

- ❖ **ProbAI** – Probabilistic AI  Trondheim, 2025
- ❖ **SIPTA** – Imprecise Probabilities  Ghent, 2024
- ❖ **ACDL** – Advanced Course on Data Science & Machine Learning  Grosetto, 2023

Service & Outreach

- ❖ Peer reviewer for ECMLPKDD, NeurIPS.
- ❖ Co-organized “Scientific Coffee” research seminar series at Fraunhofer IEE.
- ❖ Completed doctoral school courses: presentation skills, time management, personal effectiveness, Julia programming.

PERSONAL INTERESTS

Outdoor activities (hiking, running, swimming), tennis